# USRP X310 Data Recording Report

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**Abstract**

This report describes the data recording process using the USRP X310 device. This reciver is placed in a fixed position in São José dos Campos, SP, Brazil, in the GPS/GNSS Laboratory at Aeronautics Institute of Technology (ITA)[1]. The present report is updated as new recordings are made and new information is available.

## 1 Overview of the recordings

The recording is computed automatically by a Linux service which is driven by the following guidelines:

- Everyday, the program will start within the time rage [4PM, 6PM]
- It will sense whether the USRP is detectable via `uhd_find_devices --args=''addr=192.168.10.2''`.
- If not found, it will exit silently and will sense again in intervals of 15 min.
- Once found, a SigMF file will be created from the recording, with the interleaved I/Q samples + metadata, like number of channels (dual channel recording), frequency (L1 and L5), sampling rate, gain, sample type (signed 8-bit int for I and for Q, 2 bytes per complex sample), etc. Each channel has its own configuration (see the table below), so the metadata of each channel is reported separately.
  - Channel 0 (RX0): GPS L1 (1575.42 MHz), 4 MS/s, 25 dB of gain;
  - Channel 1 (RX1): GPS L5 (1176.45 MHz), 20 MS/s, 25 dB of gain.

  Records made before 17/10/2026 used 2 MS/s in both channels; records made before 19/10/2026 used signed 16-bit I/Q (4 bytes per complex sample).
- Each record lasts 5 minutes (300 s)

[1] https://github.com/ita-gnss-lab

- It will record a second time within the same range and 30 min after the first record, and then it will not record anymore for that day.
- After 15 days of records from the last report. Rubem will send an automatic message to header of the Laboratory, Prof. Dr. Felix Antreich, via WhatsApp with the summarized information from all recorded metadata;

October

| M | T | W | T | F | S | S |
|---|---|---|---|---|---|---|
| | | | 1 | 2 | 3 | 4 |
| 5 | 6 | 7 | 8 | 9 | 10 | 11 |
| 12 | 13 | 14 | 15 | 16 | 17 | 18 |
| 19 | 20 | 21 | 22 | 23 | 24 | 25 |
| 26 | 27 | 28 | 29 | 30 | 31 | |

Table 1: Recording calendar. Green: 2 recordings made; blue: 1 recording made. No color: no recording made.

# 2 Recording reports

## 2.1 Records on 08/10/2026

### 2.1.1 First record

| **Parameter** | **Channel 0** | **Channel 1** |
|---|---|---|
| Start time | 2026-10-08 22:26:37 UTC | |
| Duration | 300.0 s | |
| Sample rate | 2.000 MS/s | |
| Sample type | signed complex 16-bit samples (I+Q), little-endian | |
| Frequency | 1575.420 MHz | 1176.450 MHz |
| Gain | 20.0 dB | 20.0 dB |
| Antenna | RX1 | RX2 |
| File size | 2400.0 MB | 2400.0 MB |
| Samples written | 600,000,000 | 600,000,000 |
| Lost samples | 0 | 0 |
| Clock source | external | |
| Time source | internal | |
| Motherboard | USRP X310 | |
| Daughterboard | TwinRX RX0 | TwinRX RX1 |
| Recorder | UHD Python API | |

Table 2: Metadata for recording `x310_20261008T222630Z`, captured starting 2026-10-08T22:26:37.215640Z.

## 2.2 Records on 13/10/2026

### 2.2.1 First record

| Parameter | Channel 0 | Channel 1 |
|---|---|---|
| Start time | 2026-10-13 21:57:56 UTC | |
| Duration | 300.0 s | |
| Sample rate | 2.000 MS/s | |
| Sample type | signed complex 16-bit samples (I+Q), little-endian | |
| Frequency | 1575.420 MHz | 1176.450 MHz |
| Gain | 20.0 dB | 20.0 dB |
| Antenna | RX1 | RX2 |
| File size | 2400.0 MB | 2400.0 MB |
| Samples written | 600,000,000 | 600,000,000 |
| Lost samples | 0 | 0 |
| Clock source | external | |
| Time source | internal | |
| Motherboard | USRP X310 | |
| Daughterboard | TwinRX RX0 | TwinRX RX1 |
| Recorder | UHD Python API | |

Table 3: Metadata for recording `x310_20261013T215749Z`, captured starting 2026-10-13T21:57:56.746726Z.

### 2.2.2 Second record

| Parameter | Channel 0 | Channel 1 |
|---|---|---|
| Start time | 2026-10-13 22:08:41 UTC | |
| Duration | 300.0 s | |
| Sample rate | 2.000 MS/s | |
| Sample type | signed complex 16-bit samples (I+Q), little-endian | |
| Frequency | 1575.420 MHz | 1176.450 MHz |
| Gain | 20.0 dB | 20.0 dB |
| Antenna | RX1 | RX2 |
| File size | 2400.0 MB | 2400.0 MB |
| Samples written | 600,000,000 | 600,000,000 |
| Lost samples | 0 | 0 |
| Clock source | external | |
| Time source | internal | |
| Motherboard | USRP X310 | |
| Daughterboard | TwinRX RX0 | TwinRX RX1 |
| Recorder | UHD Python API | |

Table 4: Metadata for recording `x310_20261013T220835Z`, captured starting 2026-10-13T22:08:41.589855Z.